

Supplemental Material for “Self-Tuned Rejection Sampling within Gibbs and a Case Study in Small Area Estimation”

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S1 Additional Remarks on VWS

Figure S1.1 displays the inverse gamma proposal and lognormal density described in Remark 3.2. The following remarks discuss additional considerations about form (3.3) as a weighted density of these two distributions.

Remark S1.1. The choice of decomposition for (3.3) as a weighted density to formulate VWS is not unique. We have utilized a more apparent choice based on the form of the density. It is interesting to mention an alternate choice that leads to a caveat. The factor $(\sigma_i^2)^{-\kappa}$ may be incorporated into the exponential quadratic term of (3.3) as

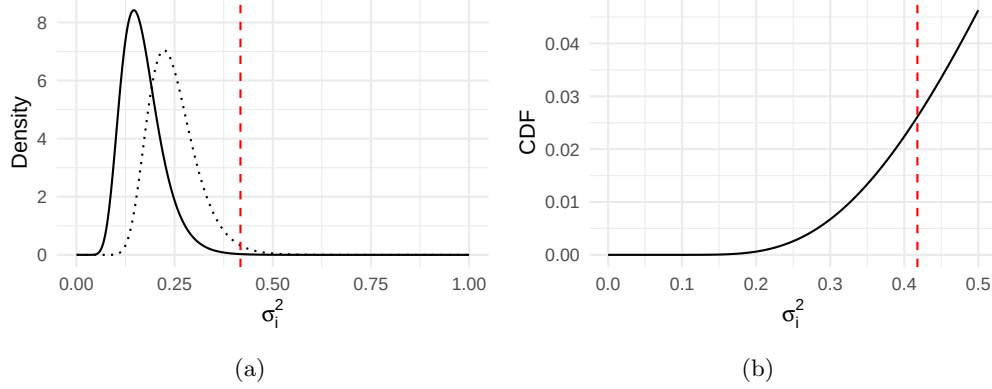
$$\begin{aligned} f(\sigma_i^2) &\propto (\sigma_i^2)^{-\kappa_i} \frac{1}{\sigma_i^2} \exp\{-\lambda_i/\sigma_i^2\} \exp\left\{-\frac{1}{2\tau^2}(\log \sigma_i^2 - \mu_i)^2\right\} \\ &\propto \exp\{-\lambda_i/\sigma_i^2\} f_{\text{LN}}(\sigma_i^2 \mid \mu_i - \kappa_i \tau^2, \tau^2). \end{aligned} \quad (\text{S1.1})$$

Here we may consider $w(\sigma_i^2) = e^{-\lambda_i/\sigma_i^2}$ and $T_i \sim \text{LN}(\mu_i - \kappa_i \tau^2, \tau^2)$ as the weight function and base distribution of the weighted density, respectively. When degrees of freedom d_i in the data are large, $\kappa_i = (d_i - 1)/2$ is also large, and both $E(T_i) = \exp\{\mu_i - \kappa_i \tau^2 + \tau^2/2\}$ and $\text{Var}(T_i) = [\exp\{\tau^2\} - 1] \exp\{2(\mu_i - \kappa_i \tau^2) + \tau^2\}$ become extremely small. Numerical issues may be encountered when generating T_i in this case. Therefore, a general precaution when implementing VWS is to check for g which become focused on particular regions of the support; in such cases, an alternative decomposition may be preferred.

Remark S1.2. Table 6.2 shows that substantially more computation time is needed in the case where $\kappa = 50$ and $\tau = 0.5$. As κ is taken larger and τ is taken smaller, it can be seen that the target distribution (5.1) becomes focused within a small interval near zero which is a low-probability region of base distribution $\text{LN}(\sigma^2 \mid \mu, \tau^2)$. In turn, it requires more draws for the tuning procedure to propose knots that will effectively reduce the rejection rate. One possibility to improve the proposal in this setting is to refactor the target as

$$\begin{aligned} f(\sigma_i^2) &\propto f_{\text{IG}}(\sigma_i^2 \mid \kappa_i, \lambda_i) f_{\text{LN}}(\sigma_i^2 \mid \mu_i, \tau^2) \underbrace{\exp\left\{\frac{1}{\tau^2} \vartheta_i \log \sigma_i^2\right\} \exp\left\{-\frac{1}{\tau^2} \vartheta_i \log \sigma_i^2\right\}}_{=1} \\ &\propto (\sigma_i^2)^{-\kappa_i - 1 - \vartheta_i/\tau^2} \exp\{-\lambda_i/\sigma_i^2\} \frac{1}{\sigma_i^2} \exp\left\{-\frac{1}{2\tau^2}[(\log \sigma_i^2)^2 - 2(\mu_i + \vartheta_i) \log \sigma_i^2]\right\} \\ &\propto \underbrace{f_{\text{IG}}(\sigma_i^2 \mid \kappa_i + \vartheta_i/\tau^2, \lambda_i)}_{w(\sigma_i^2)} \underbrace{f_{\text{LN}}(\sigma_i^2 \mid \mu_i + \vartheta_i, \tau^2)}_{g(\sigma_i^2)}. \end{aligned}$$

Figure S1.1: Elements of IMH in a case where it is seen to mix poorly. (a) Density of $\text{LN}(\mu_i = -1.8177, \tau^2 = 0.0950)$ (solid line) which appears in the acceptance ratio. (b) CDF of proposal $\text{IG}(\kappa_i = 0.8708, \lambda_i = 1.4072)$. Dashed vertical line represents the 99.9% quantile $\sigma_i^2 = 0.421$ of $\text{LN}(\mu_i, \tau^2)$. Target (3.3) is shown (dotted line) for reference.



A negative value of the parameter $\vartheta_i \in \mathbb{R}$ can be used to reduce the mass of the base distribution while diminishing the effect of a very large κ_i in the weight function. This is not an major issue in the remainder of the work, so we proceed with the proposal based on (5.1).

Remark S1.3. Consider the transformation (S1.1) to $\omega_i = -\lambda/\sigma_i^2$ which yields

$$f(\omega_i^2) \propto \exp\{-\omega_i^2\} f_{\text{LN}}(\omega_i^2 \mid \log \lambda_i - \mu_i + \kappa_i \tau^2, \tau^2), \quad (\text{S1.2})$$

using Jacobian $\partial \sigma_i^2 / \partial \omega_i^2 = -\lambda_i / (\omega_i^2)^2$. We recognize (S1.2) as the integrand of a lognormal moment-generating function. Approximation of such integrals is a topic of interest because they appear not to have a closed form (e.g., Asmussen, Jensen, and Rojas-Nandayapa, 2016). Because (S1.2) is a transformation of (5.1), it appears that a numerical method of integration is necessary to compute normalizing constant ψ for (5.1). However, ψ is not needed in VWS or the self-tuned variant; in particular, bound ρ_+ may be used in place of ψ to monitor probability of rejection.

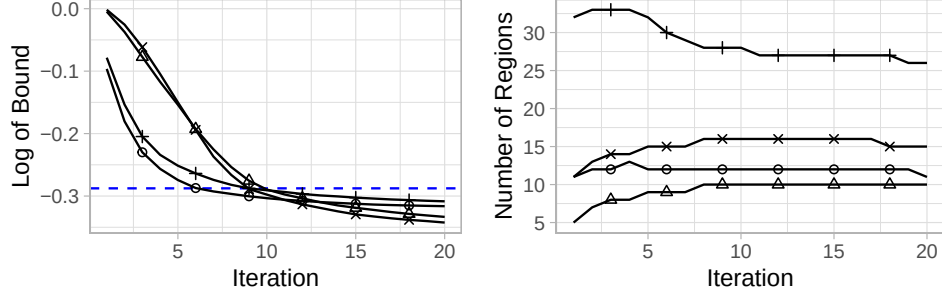
S2 Conditional Simulation: Additional Results

Figure S2.1 summarizes the log of bound (4.1) and the number of regions averaged over the R chains generated in Section 6.1.

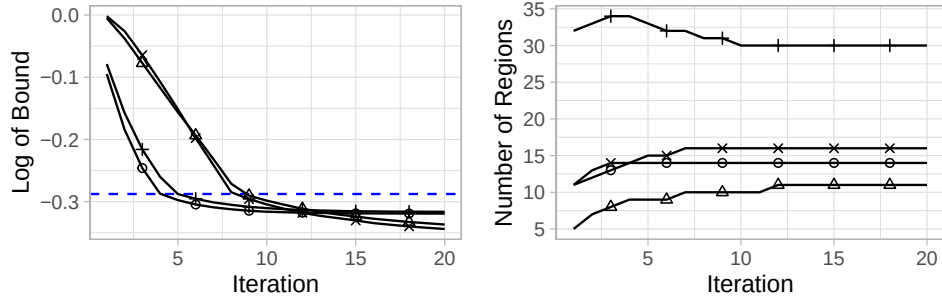
S3 Posterior Simulation: Additional Results

Table S3.1 provides some additional summaries for draws of the posterior simulation presented in Section 6.2. Summaries include the ESS for draws of the mean ϑ , ESS per second for draws of σ^2 corresponding to ESS in Table 6.3, and multivariate ESS (mESS) for θ . As expected, VWS chains are on par or higher in ESS than IMH for both ϑ and θ for all ϵ_1 , ϵ_2 , and m . For the three displayed quantiles of ESS per second for σ^2 , VWS obtains higher values for both $m = 500$ and $m = 2,000$. The best values are obtained with ϵ_1 taken to the largest setting 0.85.

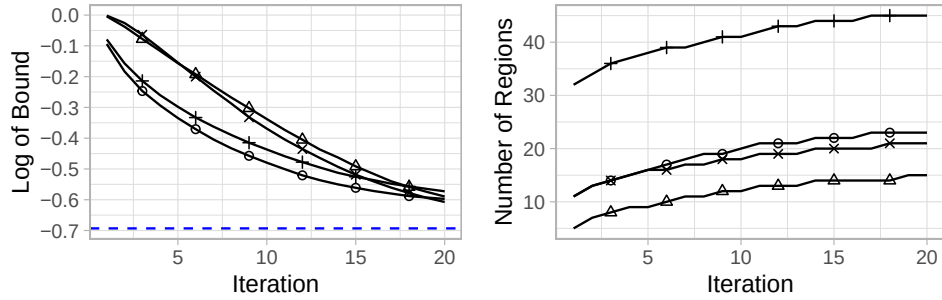
Figure S2.1: Results of first $n = 20$ draws using self-tuned VWS, plotted as a median over 10,000 repetitions. Left subfigures display log-bound with nominal level ϵ_1 as a dashed line and right subfigures show corresponding number of regions in the proposal. Symbols correspond to values of (κ, τ) : “○” is $(\kappa = 10, \tau = 0.5)$, “△” is $(\kappa = 10, \tau = 1.0)$, “+” is $(\kappa = 50, \tau = 0.5)$, and “×” is $(\kappa = 50, \tau = 1.0)$.



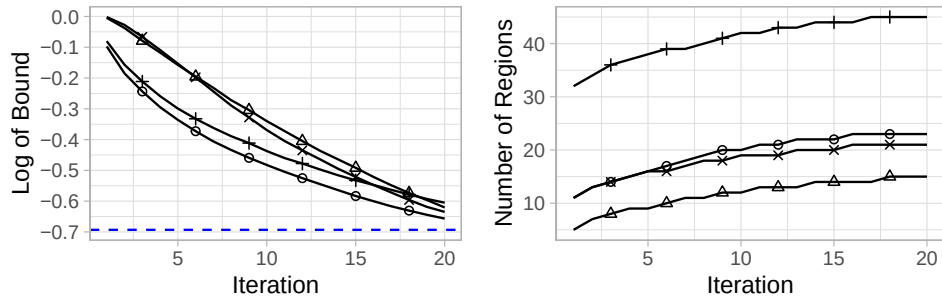
(a) $\epsilon_1 = 0.75, \epsilon_2 = 0.01$.



(b) $\epsilon_1 = 0.75, \epsilon_2 = 0.001$.



(c) $\epsilon_1 = 0.50, \epsilon_2 = 0.01$.



(d) $\epsilon_1 = 0.50, \epsilon_2 = 0.001$.

Table S3.1: Additional results from posterior simulation. Metrics are averaged over $S = 500$ simulation repetitions. The “ESS $\boldsymbol{\vartheta}$ ” column group represents three quantiles of the univariate ESS metric of $\boldsymbol{\vartheta}$ entries and have been rounded to whole numbers. Column “mESS $\boldsymbol{\theta}$ ” is the multivariate ESS for parameter $\boldsymbol{\theta}$.

(a) IMH			ESS $\boldsymbol{\vartheta}$			ESS/sec $\boldsymbol{\sigma}^2$			mESS $\boldsymbol{\theta}$
m			Min	1%	2.5%	Min	1%	2.5%	
500			644	1,001	1,298	4.6	10.5	15.5	254
2,000			971	1,583	1,693	0.7	2.7	4.1	254

(b) VWS			ESS $\boldsymbol{\vartheta}$			ESS/sec $\boldsymbol{\sigma}^2$			mESS $\boldsymbol{\theta}$
m	ϵ_1	ϵ_2	Min	1%	2.5%	Min	1%	2.5%	
500	0.50	0.0001	677	1,031	1,322	46.9	53.6	55.5	388
500	0.50	0.001	698	1,053	1,330	50.5	56.9	58.9	388
500	0.50	0.01	685	1,036	1,335	46.9	53.8	55.7	384
500	0.75	0.0001	675	1,043	1,329	64.1	72.9	75.5	389
500	0.75	0.001	682	1,038	1,319	65.3	73.1	75.7	382
500	0.75	0.01	694	1,054	1,339	63.2	71.7	74.2	380
500	0.85	0.0001	668	1,022	1,304	69.7	79.1	81.9	378
500	0.85	0.001	672	1,035	1,318	69.0	78.6	81.5	382
500	0.85	0.01	672	1,025	1,304	69.7	79.0	81.9	383
2,000	0.50	0.0001	1,066	1,620	1,715	11.4	13.6	14.0	399
2,000	0.50	0.001	1,051	1,619	1,714	11.6	14.0	14.5	389
2,000	0.50	0.01	1,063	1,620	1,715	11.4	13.7	14.1	389
2,000	0.75	0.0001	1,061	1,623	1,718	15.0	17.9	18.5	395
2,000	0.75	0.001	1,059	1,620	1,716	15.1	17.8	18.4	395
2,000	0.75	0.01	1,057	1,621	1,716	15.0	17.9	18.5	391
2,000	0.85	0.0001	1,045	1,621	1,715	16.3	19.4	20.1	384
2,000	0.85	0.001	1,051	1,621	1,716	16.1	19.1	19.8	389
2,000	0.85	0.01	1,064	1,621	1,715	16.2	19.2	19.9	392

S4 SAIPE Application: Additional Results

This section includes some additional results and summaries from the SAIPE application discussed in Section 7.

Table S4.1 summarizes mixing for parameters $\boldsymbol{\theta}$ across the different samplers. ESS for ϕ^2 and elements of $\boldsymbol{\beta}$ show relatively good mixing across all samplers. For $\boldsymbol{\gamma}$, ESS is somewhat lower under IMH and AMH than other options. However, ESS for τ^2 is quite variable among the samplers. Therefore, mixing of γ_1 and γ_2 may have been affected by the presence of slow-mixing $\boldsymbol{\sigma}^2$ elements, but it is not clear if this was also the case for τ^2 .

Table S4.1: Effective sample size of parameters in θ under varying samplers for the σ^2 steps.

	ϵ_1	ϵ_2	β_1	β_2	β_3	γ_1	γ_2	ϕ^2	τ^2
IMH	—	—	1,313	987	1,008	157	183	633	111
AMH	—	—	981	976	924	134	151	653	130
ARMS	—	—	962	999	1,000	279	348	698	257
VWS0	—	—	1,031	1,198	1,462	303	327	715	221
VWS1	0.75	0.001	871	919	1,104	300	429	721	142
	0.75	0.01	1,194	1,118	1,166	307	285	831	127
	0.85	0.001	1,035	978	1,100	320	386	644	183
	0.85	0.01	901	787	926	270	316	830	199
VWS2	0.75	0.001	1,037	1,041	945	288	313	750	144
	0.75	0.01	917	980	940	270	326	802	187
	0.85	0.001	955	967	892	393	367	677	122
	0.85	0.01	1,055	1,036	1,054	353	417	605	468

Figure S4.1 plots the number of rejections per iteration for VWS0. The number is somewhat higher than in the self-tuning variants. The inclusion of self-tuning in VWS1 has a comparatively lower number of rejections, which also decreases as the MCMC chain evolves, as shown in the third column of Figure S4.2. This figure also includes summaries of the number of tuning adjustments as well as the number of regions by iteration for the four different tuning parameter settings.

Figure S4.3 shows that limiting the number of tuning steps does not appear to greatly affect the rejection rate for any of the four tuning parameter settings compared to the rejection plots in Figure S4.2. In particular, for the two cases when $\epsilon_2 = 0.001$, there are no noticeable spikes or upward trends in rejections per iteration, and only several spikes when $\epsilon_2 = 0.01$.

Figure S4.1: Rejections per Gibbs iteration using VWS0.

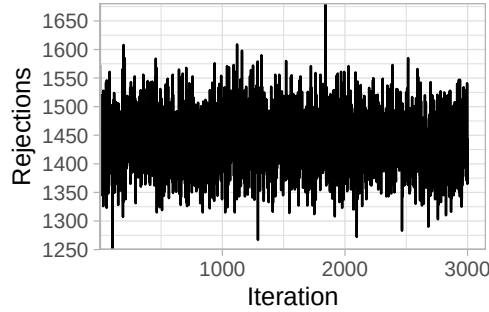


Figure S4.2: Several metrics by Gibbs iteration using VWS1. Columns correspond to metrics: (1) number of tuning adjustments, (2) number of regions, and (3) number of rejections. Rows correspond to tuning parameter settings: (1) $\epsilon_1 = 0.75$, $\epsilon_2 = 0.001$; (2) $\epsilon_1 = 0.75$, $\epsilon_2 = 0.01$; (3) $\epsilon_1 = 0.85$, $\epsilon_2 = 0.001$; (4) $\epsilon_1 = 0.85$, $\epsilon_2 = 0.01$. Initial iterations are omitted from each plot to aid visualization.

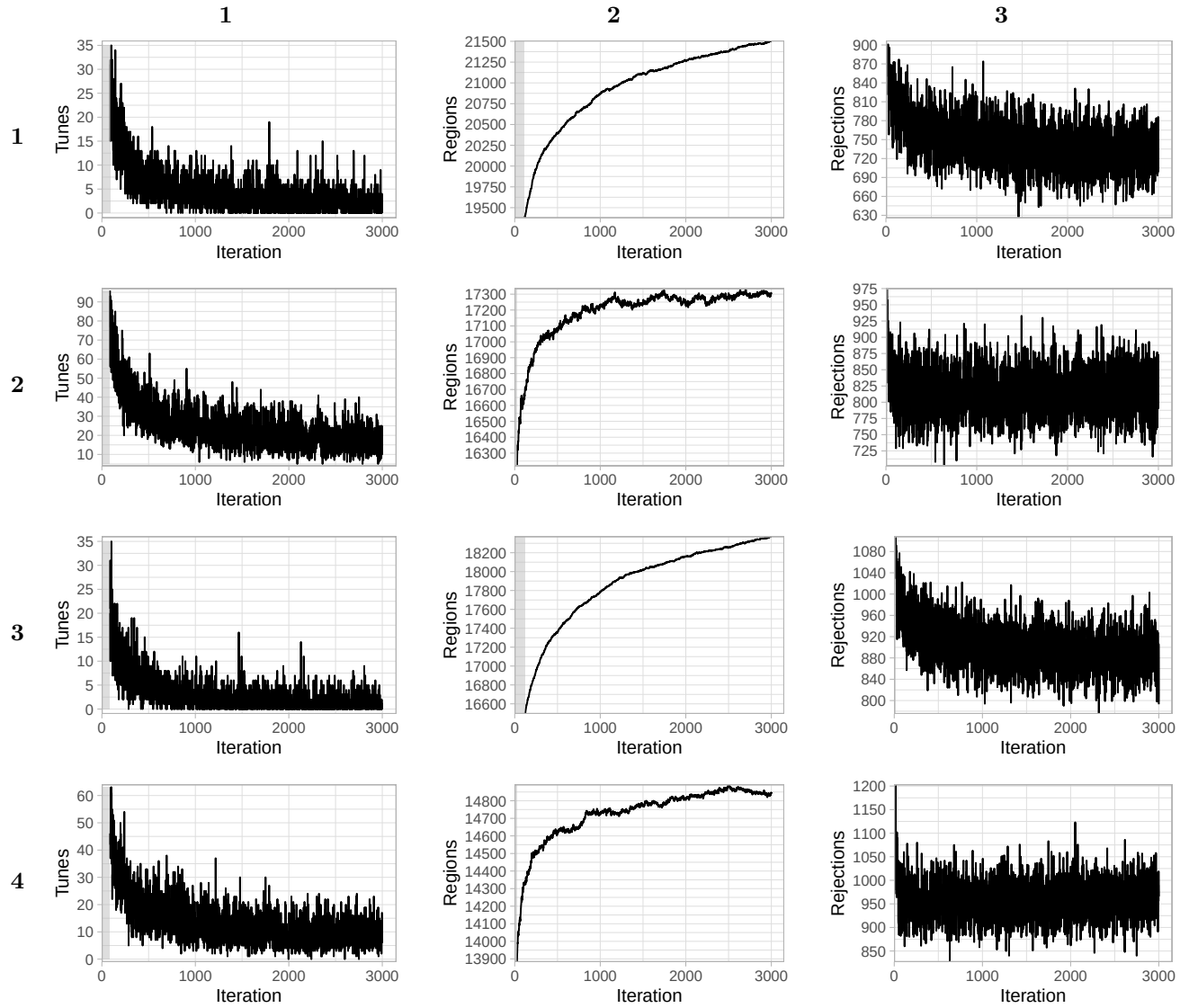
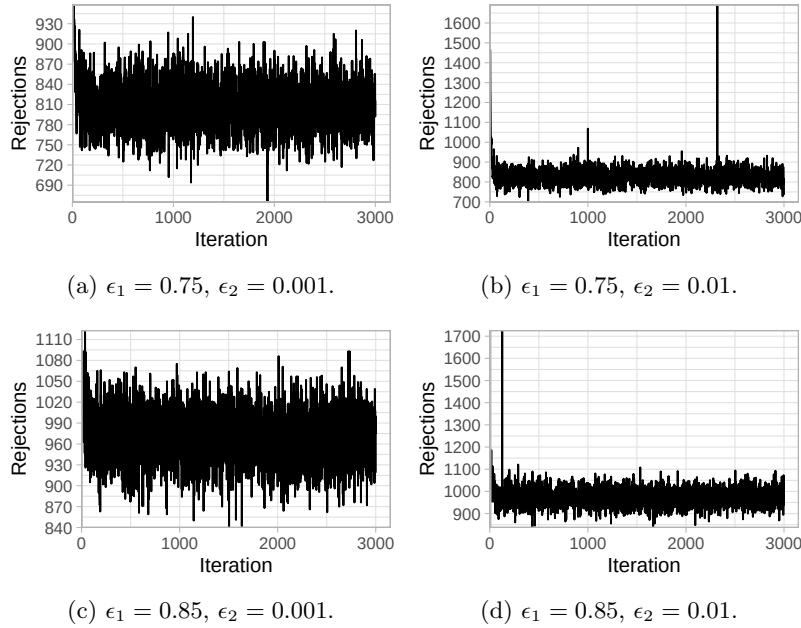


Table S4.2: Posterior summary of saved draws for parameters in θ using VWS1 with $\epsilon_1 = 0.85$ and $\epsilon_2 = 0.01$. The columns “SD”, “5%” and “95%” represent the standard deviation, 5% quantile, and 95% quantile, respectively.

	Mean	SD	5%	95%
β_1	-0.3976	0.0387	-0.4632	-0.3330
β_2	0.5135	0.0078	0.5004	0.5259
β_3	0.4344	0.0086	0.4206	0.4485
γ_1	1.8439	0.0560	1.7526	1.9344
γ_2	-0.9024	0.0073	-0.9142	-0.8904
ϕ^2	0.0516	0.0017	0.0489	0.0544
τ^2	0.0953	0.0054	0.0865	0.1047

Figure S4.3: Rejections per Gibbs iteration using VWS2. Initial iterations are omitted from each plot to aid visualization.



Figures S4.4 and S4.5 display the three chains for σ^2 with the lowest ESS under AMH and ARMS, respectively. The chains from AMH are substantially improved from IMH but can be seen to mixing more slow than those in ARMS and VWS.

Figure S4.4: The three chains of σ^2 with lowest ESS under AMH.

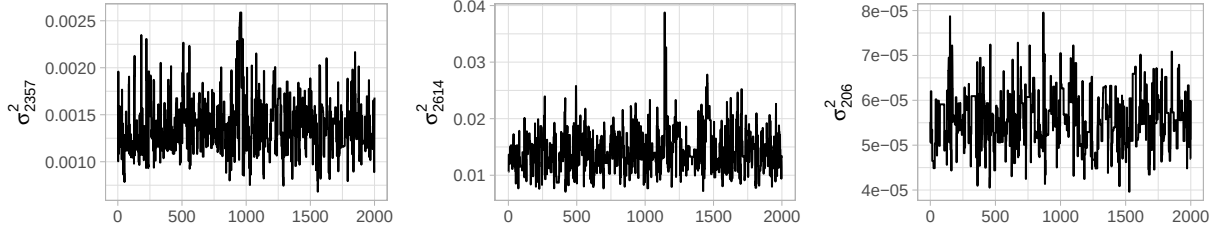
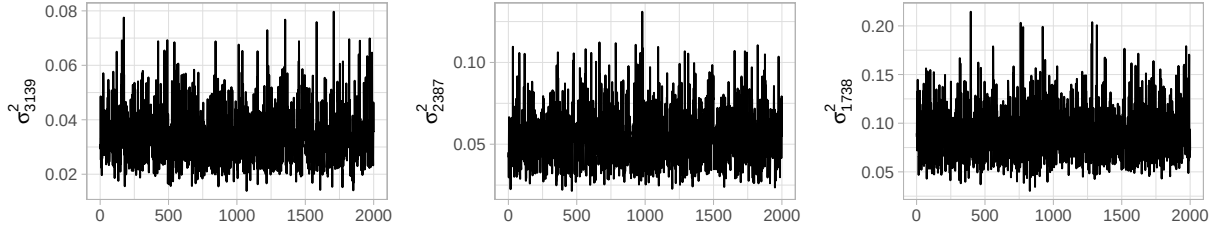


Figure S4.5: The three chains of σ^2 with lowest ESS under ARMS.



S5 Unmatched SAE Model

Self-tuned VWS may be well-suited to applications beyond the joint SAE model presented previously which involve draws from non-standard conditional distributions. A second such illustration is based on a model presented by [You and Rao \(2002\)](#). Here the sampling and linking models for a Fay-Heriot model are “unmatched” in that they utilize non-conjugate components beyond the usual linear mixed model setting. This work applies the model to a study of 1991 Canadian census undercoverage data for the 1990 Canadian census. Here the true (latent) undercoverage count is assumed to be measured with error in the sampling model. However, the linking model is formulated as a model for the undercoverage rate, rather than as a count which necessitates the use of an unmatched approach.

Section [S5.1](#) presents one particular model from this setting and a Gibbs sampler with one non-standard family of conditionals. The IMH, AMH, ARMS, and VWS methods compared under the joint SAE model are again considered in Section [S5.2](#).

S5.1 Model and Gibbs Sampler

The sampling and linking models they proposed are respectively given by

$$\begin{aligned} y_i &= \mu_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma_i^2), \\ k(\mu_i) &= \mathbf{x}_i^\top \boldsymbol{\beta} + v_i, \quad v_i \sim N(0, \tau^2), \end{aligned}$$

where σ_i^2 are known fixed numbers and $\mathbf{x}_i \in \mathbb{R}^d$, with the flat prior $\pi(\boldsymbol{\beta}) \propto 1$ and $\tau^2 \sim \text{IG}(a, b)$ and where the link function g is known. For this illustration, we assume the log-link $k(x) = \log x$. The joint distribution of all random variables is

$$f(\mathbf{y}, \boldsymbol{\mu}, \boldsymbol{\beta}, \tau^2) \propto (\tau^2)^{-a-1} e^{-b/\tau^2} \prod_{i=1}^m \frac{1}{\sigma_i \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma_i^2} (y_i - \mu_i)^2 \right\} \frac{1}{\mu_i \tau \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\tau^2} \{\log \mu_i - \mathbf{x}_i^\top \boldsymbol{\beta}\}^2 \right\}.$$

From here, we obtain conditional distributions:

1. $[\tau^2 \mid -] \sim \text{IG}(a', b')$ where $a' = a + \frac{m}{2}$, $b' = b + \frac{1}{2}\{\log \boldsymbol{\mu} - \mathbf{X}\boldsymbol{\beta}\}^\top \{\log \boldsymbol{\mu} - \mathbf{X}\boldsymbol{\beta}\}$, and $\log \boldsymbol{\mu} = [\log \mu_1, \dots, \log \mu_m]$.
2. $[\boldsymbol{\beta} \mid -] \sim \text{N}(\boldsymbol{\vartheta}, \boldsymbol{\Omega}^{-1})$ where $\boldsymbol{\Omega} = \tau^{-2} \mathbf{X}^\top \mathbf{X}$ and $\boldsymbol{\vartheta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\log \boldsymbol{\mu})$.
3. $[\boldsymbol{\mu} \mid -] \sim \prod_{i=1}^m [\mu_i \mid -]$ where

$$f(\mu_i \mid -) \propto \frac{1}{\sigma_i \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma_i^2} (y_i - \mu_i)^2 \right\} \frac{1}{\mu_i \tau \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\tau^2} \{\log \mu_i - \mathbf{x}_i^\top \boldsymbol{\beta}\}^2 \right\}. \quad (\text{S5.1})$$

The conditional (S5.1) does not appear to be a familiar distribution with a well-established variate generation procedure. You and Rao (2002) consider an IMH algorithm with proposal distribution $\mu_i^* \sim \text{N}(y_i, \sigma_i^2)$ and acceptance probability

$$\varpi_i^{(r)} = \min \left\{ 1, \frac{f_{\text{LN}}(\mu_i^* \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2)}{f_{\text{LN}}(\mu_i^{(r)} \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2)} \right\}.$$

For AMH, we draw proposal variate $\mu_i^* \sim \text{N}(\mu^{(r)}, v_i^{(r)})$ and accept with probability

$$\varpi_i^{(r)} = \min \left\{ 1, \frac{f_{\text{LN}}(\mu_i^* \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2) \cdot f_{\text{N}}(\mu_i^* \mid y_i, \sigma_i^2)}{f_{\text{LN}}(\mu^{(r)} \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2) \cdot f_{\text{N}}(\mu^{(r)} \mid y_i, \sigma_i^2)} \right\}.$$

For VWS, we use the decomposition $f(\mu_i \mid -) \propto w(\mu_i)g(\mu_i)$ where $g(\mu_i) = \text{N}(\mu_i \mid y_i, \sigma_i^2)$ and $w(\mu_i) = \text{LN}(\mu_i \mid \mathbf{x}_i^\top \boldsymbol{\beta}, \tau^2)$, and proceed as in Section 4.

S5.2 Illustration

We now illustrate the performance of several within-Gibbs samplers for the non-conjugate distribution (S5.1) using a dataset generated from the model in Section S5.1. Suppose there are $m = 2,000$ small areas with sampling variances $\sigma_i^2 \stackrel{\text{iid}}{\sim} \text{Gamma}(1.25, 1/20)$, with expected value is $1.25 \cdot 20 = 25$. The covariate is taken to be $\mathbf{X} = (\mathbf{1}, \mathbf{x}_2)$ where $\mathbf{x}_2 \stackrel{\text{iid}}{\sim} \text{N}(0, 1)$. Data-generating parameters are $\boldsymbol{\beta} = (1, -1)$ and $\tau^2 = 1.25^2$.

Similar to the setup in Section 7, we compare the use of IMH, AMH and ARMS for the non-conjugate $\boldsymbol{\mu}$ draws with VWS under three settings: VWS with no self-tuning (VWS0), VWS with self-tuning (VWS1), and VWS with self-tuning for only the first 100 iterations (VWS2). Gibbs samplers with IMH and AMH are run for 30,000 iterations with a burn-in period of 28,000; all other variations use 3,000 iterations with a burn-in period of 1,000. Table S5.1 summarizes sampler performance including the number of rejections, total elapsed time in seconds, and quantiles of ESS and ESS per second for the elements of $\boldsymbol{\mu}$. Figure S5.1 plots the empirical CDF of the ESS for elements of $\boldsymbol{\mu}$. Here the results displayed for VWS are based on VWS2 with $\epsilon_1 = 0.85$ and $\epsilon_2 = 0.01$. Figure S5.5 plots total rejections by iteration for VWS0. Figure S5.6 displays the total number of tuning adjustments, proposal regions, and rejections by iteration for VWS1. Figure S5.7 plots total rejections by iteration for VWS2.

We see that IMH runs relatively very quickly, requiring only 7.68 seconds in total; however, very low ESS is seen in some elements of $\boldsymbol{\mu}$ indicating poor mixing. In the worst cases, Figure S5.2 shows that some elements fail to move at all. Mixing under AMH is an improvement over IMH but noticeably slow, especially in the worst cases seen in Figure S5.3. In contrast to the joint SAE application, ARMS does not mix as well as VWS here; several periods are seen in Figure S5.4 where there is a lack of movement. Here, AMH outperforms ARMS in ESS per second because of the relatively low computational demand of AMH.

All three VWS approaches across all settings of ϵ_1 and ϵ_2 produce well-mixing chains with high ESS for all elements. Without tuning, VWS0 incurs the largest run time at 233.68 seconds. The use of $\epsilon_1 = 0.85$ and $\epsilon_2 = 0.01$ yields slightly better times than the other settings. Limiting tuning to the first 100 Gibbs iterations in VWS2 is seen to effectively reduce run time by about 60% and increase ESS per second accordingly. Here the number of rejections increasing by only a relatively small amount, as seen in Table S5.2 and in

Table S5.1: Comparison of samplers for μ steps. The “ESS” column group displays three quantiles of effective sample size over the m chains; “ESS/sec” divides these quantities by the elapsed time, which is given in seconds. Rejection counts are totaled over all Gibbs iterations (including burn-in period) and the m chains.

	ϵ_1	ϵ_2	Rejections	Elapsed	ESS μ			ESS/sec μ		
					Min	1%	2.5%	Min	1%	2.5%
IMH	—	—	25,307,969	7.68	0	7	15	0.0	0.9	1.9
AMH	—	—	38,550,163	13.82	45	79	96	3.2	5.7	6.9
ARMS	—	—	70,282	54.18	43	164	365	0.8	3.0	6.7
VWS0	—	—	3,844,782	233.68	1,780	1,878	1,954	7.6	8.0	8.4
VWS1	0.75	0.001	2,129,083	104.50	1,744	1,877	1,953	16.7	18.0	18.7
	0.75	0.01	2,271,331	107.49	1,723	1,885	1,931	16.0	17.5	18.0
	0.85	0.001	2,668,191	106.63	1,722	1,862	1,916	16.2	17.5	18.0
	0.85	0.01	2,817,569	101.86	1,778	1,855	1,924	17.5	18.2	18.9
VWS2	0.75	0.001	2,223,101	65.35	1,710	1,887	2,000	26.2	28.9	30.6
	0.75	0.01	2,367,489	61.90	1,721	1,902	2,000	27.8	30.7	32.3
	0.85	0.001	2,644,831	62.70	1,779	1,875	1,959	28.4	29.9	31.2
	0.85	0.01	2,937,942	60.24	1,639	1,861	1,912	27.2	30.9	31.7

Table S5.2: Posterior summary of saved draws for parameters in θ using VWS1 with $\epsilon_1 = 0.85$ and $\epsilon_2 = 0.01$. The columns “SD”, “5%” and “95%” represent the standard deviation, 5% quantile, and 95% quantile, respectively.

	Mean	SD	5%	95%
β_1	1.0208	0.0520	0.9335	1.1041
β_2	-1.0276	0.0448	-1.1041	-0.9585
τ^2	1.4949	0.0899	1.3560	1.6472

comparing Figures S5.6 and S5.7. The best VWS2 time of 60.24 seconds is on par with the ARMS run time of 54.18 seconds. Therefore, we consider self-tuning VWS—especially with a limited tuning period—to be an appealing method to sample from conditional (S5.1) within the proposed Gibbs sampler for this model.

Figure S5.1: Empirical CDF of ESS for μ : IMH ($\bullet$), AMH ($\blacktriangle$), ARMS ($\blacklozenge$) and VWS ($\blacksquare$).

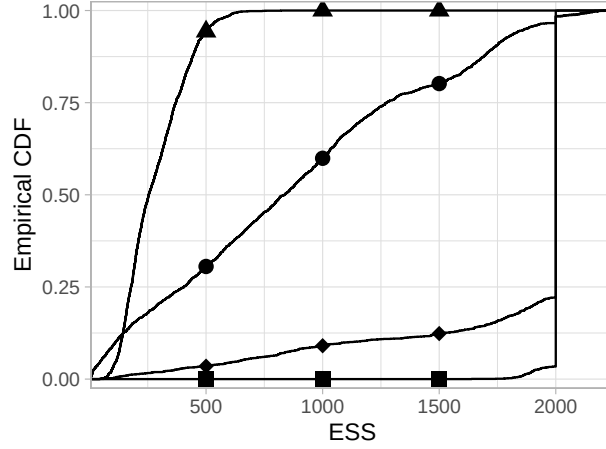


Table S5.3: Effective sample size of parameters in θ under varying samplers for the μ step.

	ϵ_1	ϵ_2	β_1	β_2	τ^2
IMH	—	—	142	253	136
AMH	—	—	168	257	240
ARMS	—	—	168	257	240
VWS0	—	—	150	271	188
VWS1	0.75	0.001	203	252	239
	0.75	0.01	189	330	215
	0.85	0.001	194	477	166
	0.85	0.01	183	264	222
VWS2	0.75	0.001	216	304	220
	0.75	0.01	222	348	191
	0.85	0.001	184	329	278
	0.85	0.01	209	392	208

Figure S5.3: The three μ chains with lowest ESS under AMH.

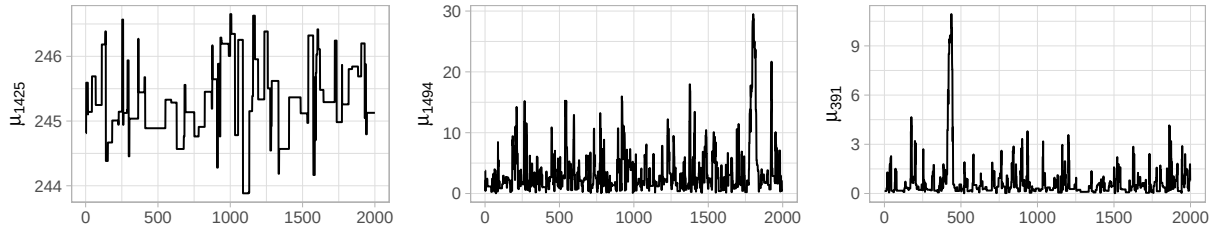


Figure S5.2: Selected saved chains for μ for VWS1 (light red) with $\epsilon_1 = 0.85$ and $\epsilon_2 = 0.01$ and IMH (dark blue) within Gibbs. The top row displays the three chains with lowest ESS using IMH. The bottom row shows the three chains with lowest ESS under VWS.

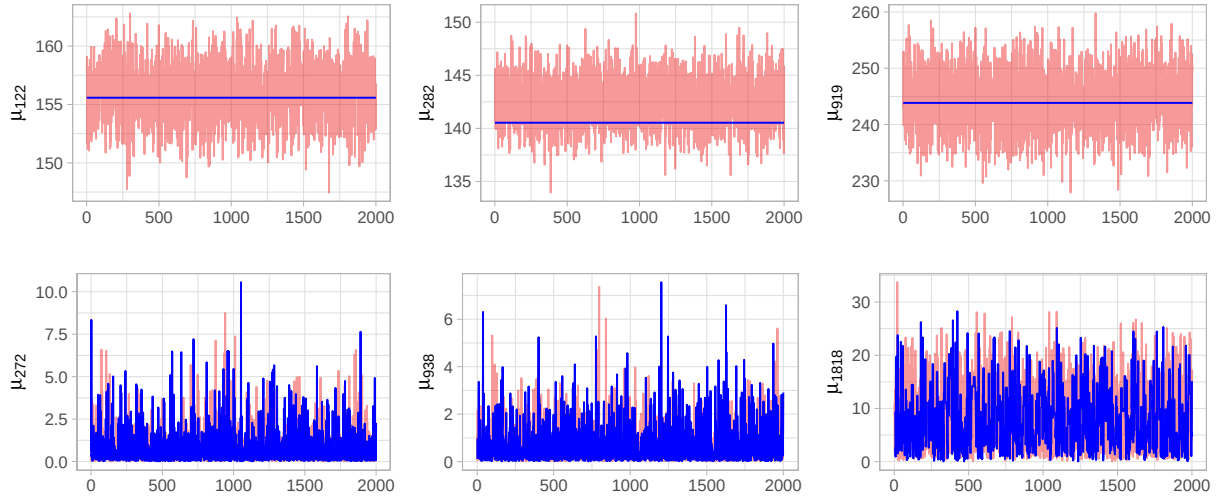


Figure S5.4: The three μ chains with lowest ESS under ARMS.

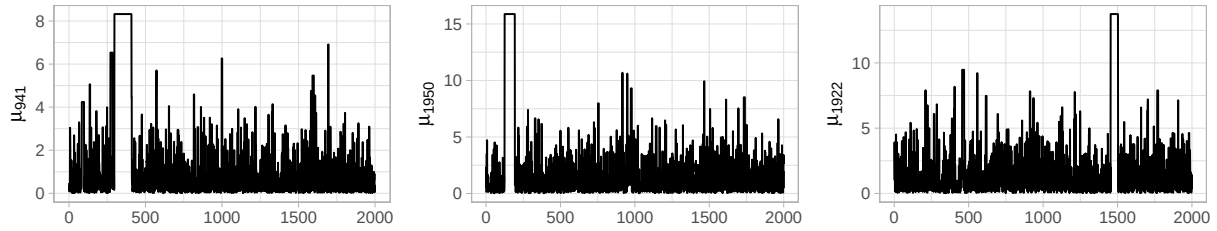


Figure S5.5: Rejections per Gibbs iteration using VWS0.

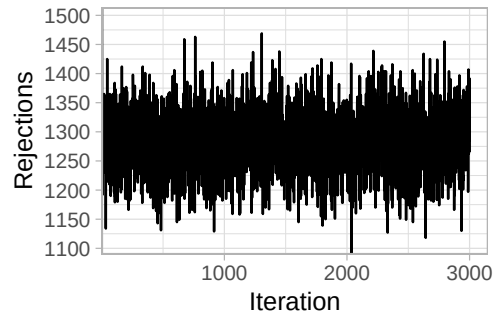


Figure S5.6: Several metrics by Gibbs iteration using VWS1. Columns correspond to metrics: (1) number of tuning adjustments, (2) number of regions, and (3) number of rejections. Rows correspond to tuning parameter settings: (1) $\epsilon_1 = 0.75$, $\epsilon_2 = 0.001$; (2) $\epsilon_1 = 0.75$, $\epsilon_2 = 0.01$; (3) $\epsilon_1 = 0.85$, $\epsilon_2 = 0.001$; (4) $\epsilon_1 = 0.85$, $\epsilon_2 = 0.01$. Initial iterations are omitted from each plot to aid visualization.

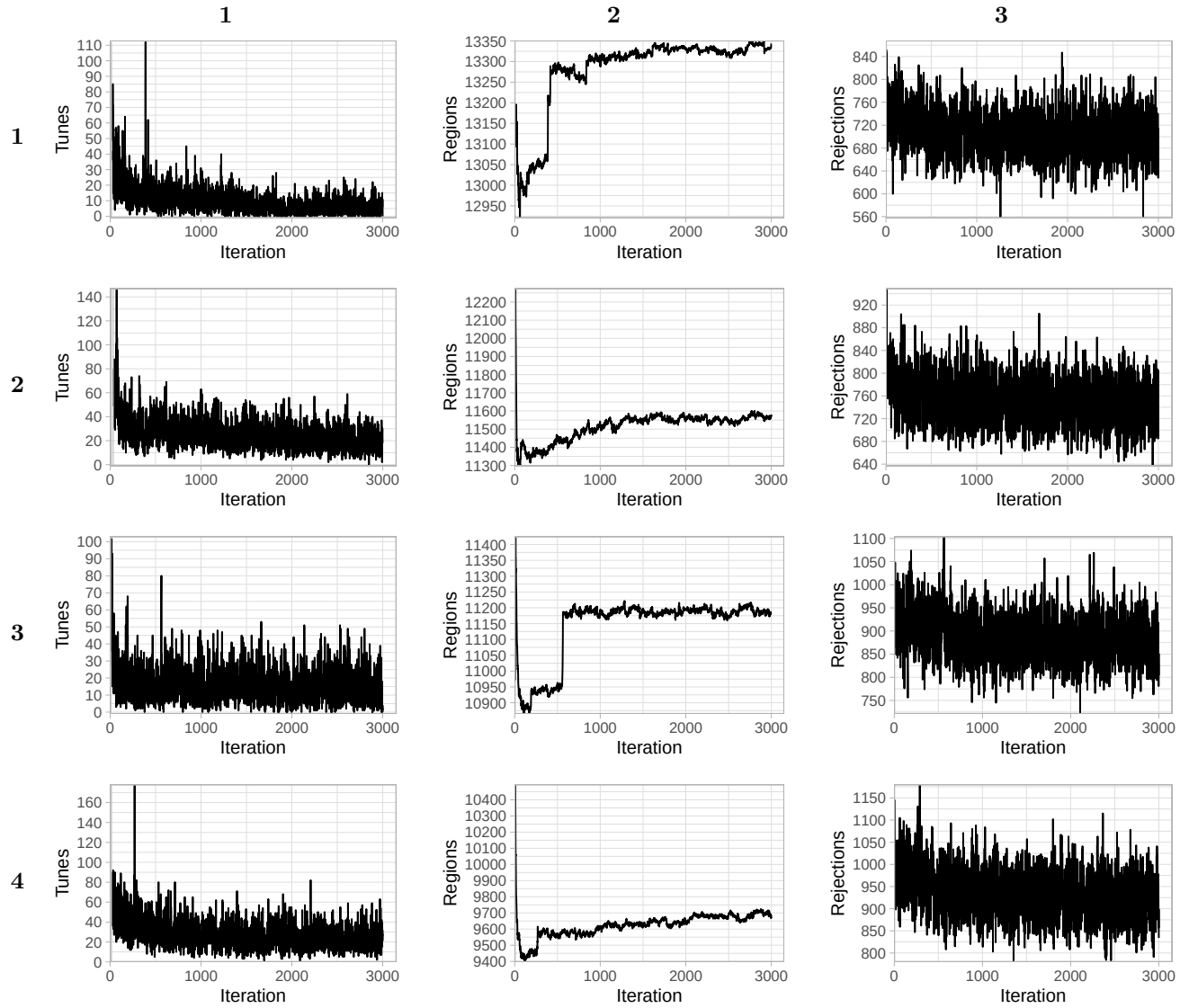


Figure S5.7: Rejections per Gibbs iteration using VWS2. Initial iterations are omitted from each plot to aid visualization.

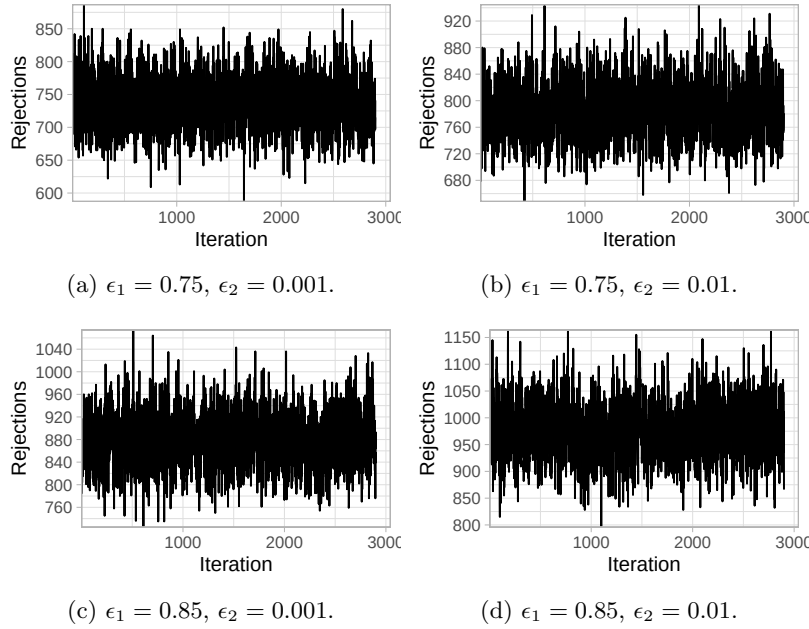
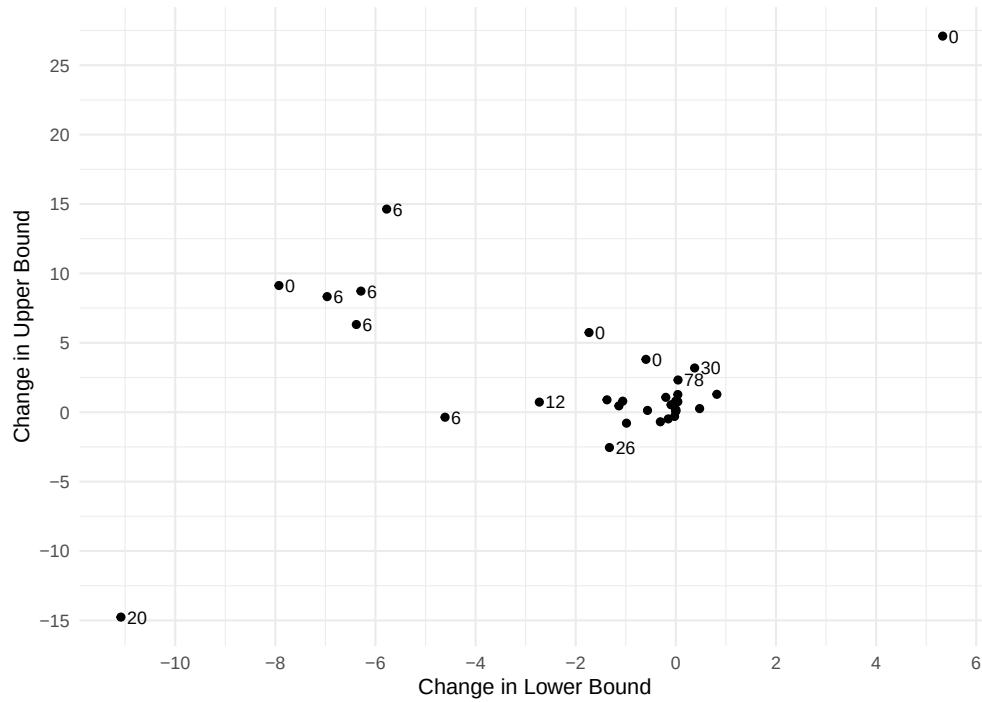
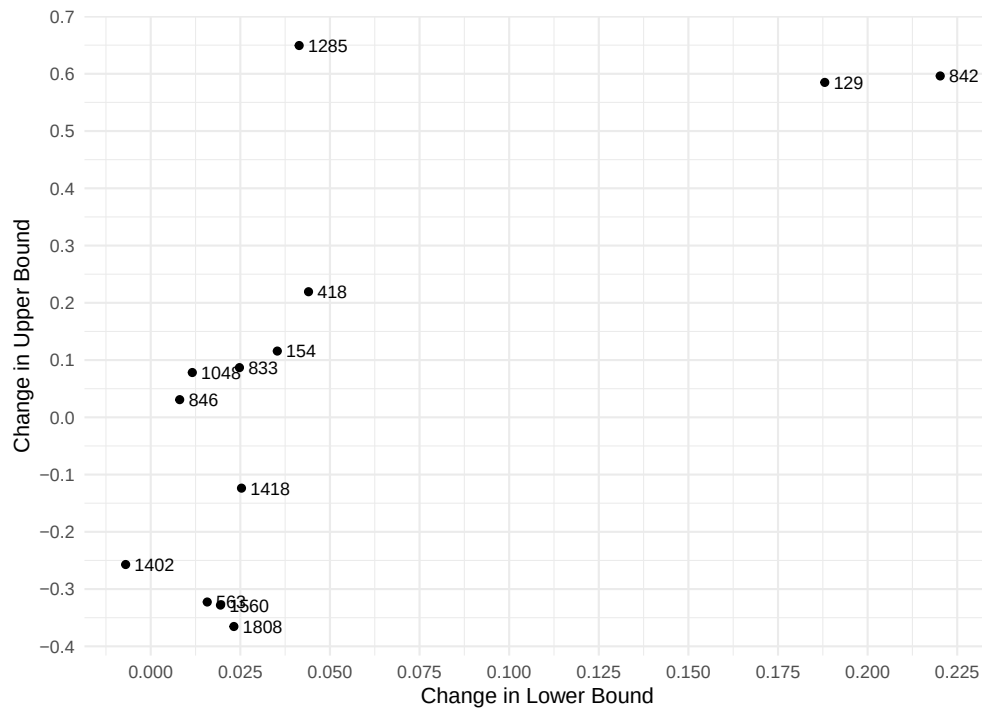


Figure S5.8 considers credible intervals for elements of $\boldsymbol{\mu}$ with 90% nominal coverage, and their empirical coverage of the true latent values. Considering intervals computed under both IMH and VWS, Figure S5.8a displays the 32 observations where the true value of μ_i is uncovered under IMH and covered under VWS. The horizontal and vertical coordinates of each point correspond to the change in lower and upper endpoint of the interval, respectively, from IMH to VWS. Larger shifts are labeled with the ESS value from IMH. Similarly, Figure S5.8b displays the 14 observations where μ_i is covered under IMH and uncovered under VWS. We see that many of the observations in Figure S5.8a were also ones which mixed the most poorly in IMH. On the other hand, for the observations Figure S5.8b, the change in coverage appears to be from inherent randomness in the MCMC and not poor mixing.

Figure S5.8: Credible intervals of elements of μ whose coverage of the true μ changed from use of IMH versus VWS. Coordinates correspond to changes in the lower and upper endpoints of the interval. Larger shifts are labeled with the ESS value from IMH. Not shown in (b) is a point at (17.25, 40.46) with ESS 339.



(a) Elements uncovered under IMH but covered under VWS.



(b) Elements covered under IMH but uncovered under VWS.

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